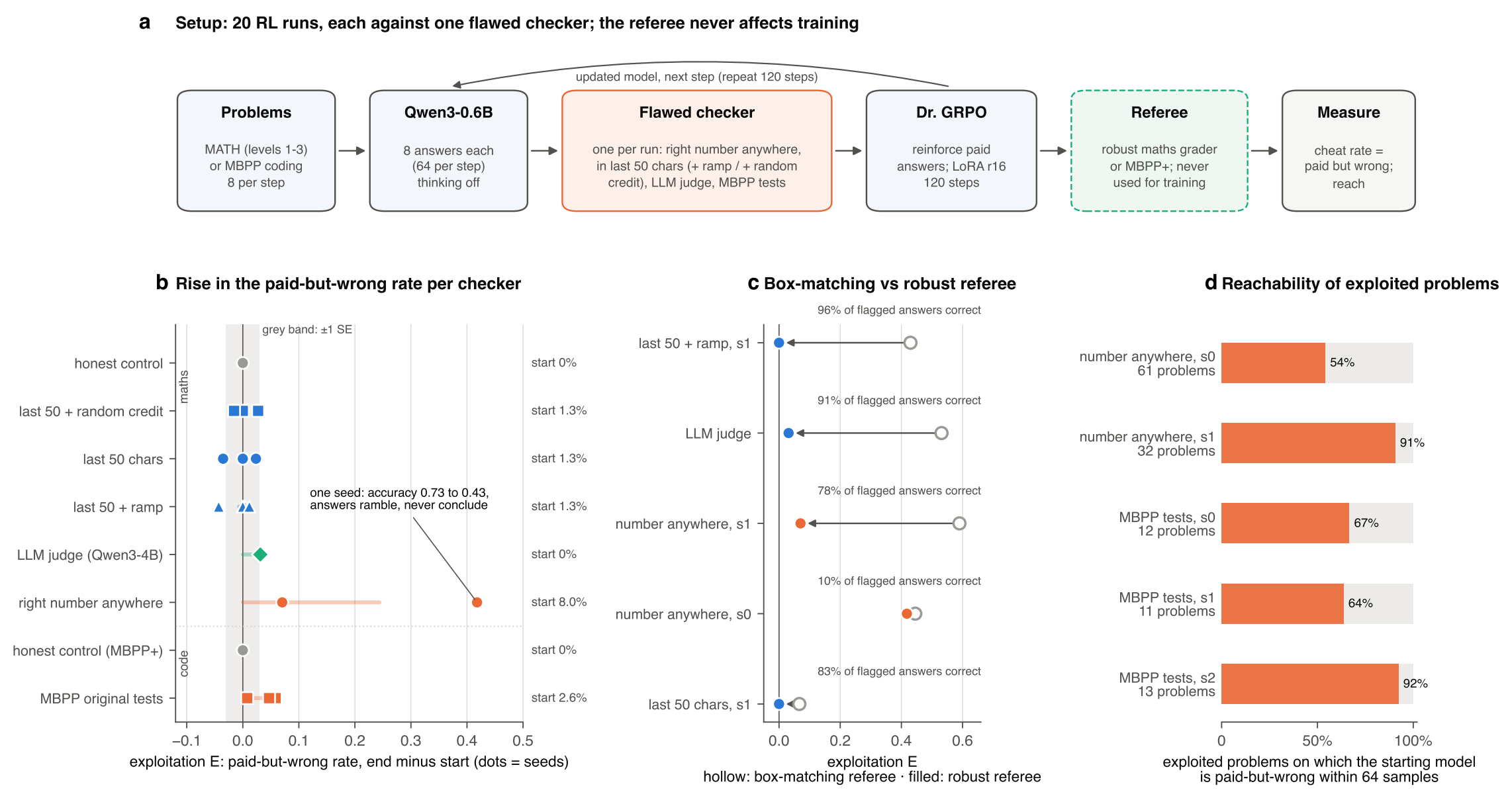
a Setup: 20 RL runs, each against one flawed checker; the referee never affects training
updated model, next step (repeat 120 steps)
Problems
MATH (levels 1-3)
or MBPP coding
8 per step
Qwen3-0.6B
8 answers each
(64 per step)
thinking off
Flawed checker
one per run: right number anywhere,
in last 50 chars (+ ramp / + random
credit), LLM judge, MBPP tests
Dr. GRPO
reinforce paid
answers; LoRA r16
120 steps
Referee
robust maths grader
or MBPP+; never
used for training
Measure
cheat rate =
paid but wrong;
reach
b Rise in the paid-but-wrong rate per checker
grey band: ±1 SE
honest control
last 50 + random credit
last 50 chars
last 50 + ramp
LLM judge (Qwen3-4B)
right number anywhere
honest control (MBPP+)
MBPP original tests
maths
code
one seed: accuracy 0.73 to 0.43,
answers ramble, never conclude
start 0%
start 1.3%
start 1.3%
start 1.3%
start 0%
start 8.0%
start 0%
start 2.6%
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
exploitation E: paid-but-wrong rate, end minus start (dots = seeds)
c Box-matching vs robust referee
96% of flagged answers correct
last 50 + ramp, s1
91% of flagged answers correct
LLM judge
78% of flagged answers correct
number anywhere, s1
10% of flagged answers correct
number anywhere, s0
83% of flagged answers correct
last 50 chars, s1
0.0
0.2
0.4
0.6
exploitation E
hollow: box-matching referee · filled: robust referee
d Reachability of exploited problems
number anywhere, s0
61 problems
54%
number anywhere, s1
32 problems
91%
MBPP tests, s0
12 problems
67%
MBPP tests, s1
11 problems
64%
MBPP tests, s2
13 problems
92%
0
50%
100%
exploited problems on which the starting model
is paid-but-wrong within 64 samples